The Cross Product

Warm-up. Let $\mathbf{u} = \langle 1, 2, 3\rangle$ and $\mathbf{v} = \langle -1, 1, 0\rangle$. (a) Compute $\mathbf{u} \cdot \mathbf{v}$. Are $\mathbf{u}$ and $\mathbf{v}$ orthogonal? (b) Can you find a nonzero vector orthogonal to *both* $\mathbf{u}$ and $\mathbf{v}$? Don't worry if (b) seems hard — that's exactly the problem we solve today.

1. Definition and computation

The dot product multiplies two vectors and produces a *number*. We now build a second kind of vector multiplication — one whose output is again a *vector*, and which happens to solve the warm-up: it manufactures a vector orthogonal to both of the ones you started with.

Definition: the cross product

The **cross product** of $\mathbf{u} = \langle u_1, u_2, u_3\rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3\rangle$ is the vector

Unlike the dot product, $\mathbf{u} \times \mathbf{v}$ is defined only when $\mathbf{u}$ and $\mathbf{v}$ are *three*-dimensional — there is no cross product of planar vectors.

Watch the output. Dotting two vectors gives a number. Crossing two vectors gives a vector. For that reason the cross product is also called the *vector product*.

A shortcut: the determinant mnemonic. The formula above is painful to memorize. Instead, write $\mathbf{i}, \mathbf{j}, \mathbf{k}$ across the top row of a symbolic 3×3 determinant, $\mathbf{u}$'s components in the second row, and $\mathbf{v}$'s components in the third; then expand along the top row.

Example 1. (a) Use the determinant shortcut to compute $\mathbf{u} \times \mathbf{v}$ for $\mathbf{u} = \langle 1, 3, 4\rangle$, $\mathbf{v} = \langle 2, 7, -5\rangle$. (b) Show that $\mathbf{u} \times \mathbf{u} = \mathbf{0}$ for *any* vector $\mathbf{u}$.

2. Magnitude and direction

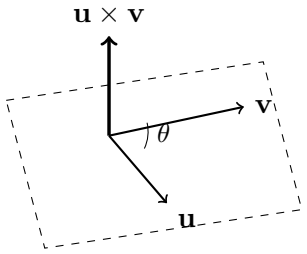
Like the dot product, the cross product has a geometric description in terms of the angle θ between $\mathbf{u}$ and $\mathbf{v}$ — but now we need *two* facts: how long $\mathbf{u} \times \mathbf{v}$ is, and which way it points.

Theorem: magnitude of the cross product

If θ is the angle between the nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ $(0 \le \theta \le \pi)$, then

Geometrically, this magnitude is the area of the parallelogram spanned by $\mathbf{u}$ and $\mathbf{v}$ (more on this in Section 4).

The direction: the right-hand rule. Place $\mathbf{u}$ and $\mathbf{v}$ in standard position. Curl the fingers of your right hand from $\mathbf{u}$ toward $\mathbf{v}$, through the angle less than 180°; your thumb then points in the direction of $\mathbf{u} \times \mathbf{v}$. In particular, $\mathbf{u} \times \mathbf{v}$ is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$, and the triple $\{\mathbf{u}, \mathbf{v}, \mathbf{u} \times \mathbf{v}\}$ is called a **right-handed system**.



Theorem: parallel vectors

Two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are **parallel** if and only if

Convention. By convention $\mathbf{0}$ is considered parallel to every vector — compare this with $\mathbf{0}$ being orthogonal to every vector for the dot product.

Example 2. (a) Are $\mathbf{u} = \langle 1, -2, 3 \rangle$ and $\mathbf{v} = \langle -2, 4, -6 \rangle$ parallel? (b) Find a vector orthogonal to both $\mathbf{p} = \langle 1, 0, -1 \rangle$ and $\mathbf{q} = \langle 2, 1, 3 \rangle$. (c) How many vectors satisfy part (b)?

Practice 1.
Let $\mathbf{v} = \langle 1, 2, -1 \rangle$ and $\mathbf{w} = \langle 3, 1, 2 \rangle$. Find $\mathbf{v} \times \mathbf{w}$. Are $\mathbf{v}$ and $\mathbf{w}$ parallel?

3. Properties of the cross product

Properties of the cross product

Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors and let c be a scalar.

Anticommutative	Distributive
Scalar multiplication	Cross product with itself
Zero property	

Contrast the middle-right entry with the dot product: $\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2$, but $\mathbf{v} \times \mathbf{v} = \mathbf{0}$.

Careful — not associative. Unlike ordinary multiplication, the cross product does *not* satisfy $(\mathbf{u} \times \mathbf{v}) \times \mathbf{w} = \mathbf{u} \times (\mathbf{v} \times \mathbf{w})$ in general. Check it yourself with the standard basis vectors: compute $(\mathbf{i} \times \mathbf{i}) \times \mathbf{j}$ and $\mathbf{i} \times (\mathbf{i} \times \mathbf{j})$

and compare.

For reference: $\mathbf{i} \times \mathbf{j} = \mathbf{k}$, $\mathbf{j} \times \mathbf{k} = \mathbf{i}$, $\mathbf{k} \times \mathbf{i} = \mathbf{j}$ — and reversing the order in any of these flips the sign.

4. Area of a parallelogram and a triangle

The magnitude theorem from Section 2 says $\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \, \|\mathbf{v}\| \sin\theta$. If $\mathbf{u}$ and $\mathbf{v}$ are adjacent sides of a parallelogram, that is exactly base times height.

Theorem: area via the cross product

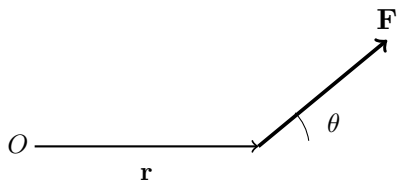
If $\mathbf{u}$ and $\mathbf{v}$ span a parallelogram, its area is

and the triangle spanned by $\mathbf{u}$ and $\mathbf{v}$ — half the parallelogram — has area

Example 3. Let $P = (1, 0, 0)$, $Q = (0, 1, 0)$, and $R = (0, 0, 1)$. (a) Find a vector orthogonal to the plane through P, Q, and R. (b) Find the area of $\triangle PQR$.

5. Torque

Turning a wrench or a screwdriver creates a rotational effect called **torque**. The farther from the axis you push, and the harder you push, the more the object turns — but only the component of the force perpendicular to the handle actually contributes.



Definition: torque

If a force $\mathbf{F}$ is applied at the point with position vector $\mathbf{r}$ (measured from the axis of rotation), the **torque** is

Units: newton-meters (N·m) or foot-pounds (ft·lb) — dimensionally the same as work, but torque is a *vector* that measures turning effect, not energy transferred.

Example 4. (a) A bolt is tightened by applying a 50-N force to a 0.2-m wrench, at an angle of 60° between the wrench and the force. Find the magnitude of the torque about the bolt. (b) A wrench is represented by $\mathbf{r} = \langle 3, 0, 0\rangle$ ft, with a force $\mathbf{F} = \langle 0, -4, 3\rangle$ lb applied at its end. Find the torque vector $\tau = \mathbf{r} \times \mathbf{F}$ and its magnitude.

Practice 2.
Find the area of the parallelogram with adjacent sides $\mathbf{u} = \langle 1, 1, 0\rangle$ and $\mathbf{v} = \langle 0, 1, 1\rangle$.

Answers to Practice. **1.** $\mathbf{v} \times \mathbf{w} = \langle 5, -5, -5\rangle \neq \mathbf{0}$; not parallel. **2.** $\mathbf{u} \times \mathbf{v} = \langle 1, -1, 1\rangle$, so the area is $\sqrt{3}$.

More practice. OpenStax §2.4: #183–190 (computation), #199–206 (orthogonal/parallel vectors), #209–212 (areas), #235 (torque).